LINEAR TRANSFORMATION

→ A transformation from a vector space V to a vectorspace W is a role that assigns each vector $x \in V$. A vector $T(x) \in W$.

→ It is written as $T: V \to W$

NOTE:

• Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$ then we have the transformation from $T: R^3 \to R^2$

ex: Let $T: \mathbb{R}^2 \to \mathbb{R}^3$ is defined by $T(x,y) = (x^2, y^2, xy)$
Then $T(2,4) = (4,16,8)$.

The set V is called domain and W is called co-domain of T.

Linear transformation:

→ Let V and W be any two vector spaces. a linear transformation $T: V \to W$ is a role that assigns each vector $u \in V$ to a unit vector.

$T(u) \in W$ such that it satisfies following

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m a vector space V to
a role that assigns
vector $T(x) \in W$.
$\rightarrow W$

then we have the
$T: R^3 \rightarrow R^2$

d by $T(x,y) = (x^2, y^2, xy)$

n and W is called

vector spaces.
$V \rightarrow W$ is a role
$\in V$ to a unit

:s following

conditions.

1. $T(U+V) = T(U) + T(V)$
2. $T(cU) = cT(U)$

$\forall\ U, V \in V$ and c be any scalar

I $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x) = 1.5x$

$\Rightarrow$ i] $T(U+V) = 1.5(U+V)$
$= 1.5U + 1.5V$
$= T(U) + T(V)$

ii] $T(cU) = 1.5(cU)$
$= c(1.5U)$
$= cT(U)$

II Define $T: R \rightarrow R$ by $T(x) = x+1$

$\Rightarrow$ $T(U+V) \neq x+1$ when $x = U$
$T(U) = U+1$

when $x = V$
$T(V) = V+1$

$T(U) + T(V) = U+1+V+1 = U+V+2$

$T(U+V) = U+V+1$

$\therefore$ $T(U+V) \neq T(U) + T(V)$

NOTE:

→ If T is a linear transformation then $T(0) = 0$

→ $T(\alpha U + \beta V) = \alpha T(U) + \beta T(V)$

let α and β be the scalars and U, V be the vectors.

1. let $T: V_3(R) \longrightarrow V_2(R)$ defined by $T(x, y, z) = (x+y, y+z)$. show that T is LT from $V_3(R) \longrightarrow V_2(R)$.

⇒ let $U = (x_1, y_1, z_1)$ and $V = (x_2, y_2, z_2)$

$T(U) = (x_1+y_1, y_1+z_1)$ and (x_2+y_2, y_2+z_2)

To prove $T(U+V) = T(U) + T(V)$

$U+V = (x_1, y_1, z_1) + (x_2, y_2, z_2)$

$= (x_1+x_2, y_1+y_2, z_1+z_2)$

$T(U+V) = T(x_1+x_2, y_1+y_2, z_1+z_2)$

$= [(x_1+x_2) + (y_1+y_2), (y_1+y_2) + (z_1+z_2)]$

$= [(x_1+y_1) + (x_2+y_2), (y_1+z_1) + (y_2+z_2)]$

$= \{[(x_1+y_1)(y_1+z_1)] + [(x_2+y_2), (y_2+z_2)]\}$

$= T(u) + T(v)$ //

$T(u) + T(v) = (x_1 + y_1, y_1 + z_1) + (x_2 + y_2, y_2 + z_2)$

$= (x_1 + y_1 + x_2 + y_2, y_1 + z_1 + y_2 + z_2)$ //

To prove $T(cu) = cT(u)$

$u = (x_1, y_1, z_1)$

$cu = c(cx_1, cy_1, cz_1)$

$T(cu) = T(cx_1, cy_1, cz_1)$

$= (cx_1 + cy_1, cy_1 + cz_1)$

$= \{c(x_1 + y_1), c(y_1 + z_1)\}$

$= \{c(x_1 + y_1), (y_1 + z_1)\}$

$= cT(u)$ //

2. Let $T: V_3 \to V_2$ defined by $T(x_1, x_2, x_3) = (x_1 - x_2, x_1 + x_3)$

Show that T is an Linear Transformation.

=> Let $(u, v) \in V_3$

$u = (x_1, y_1, z_1)$ $\quad v = (x_2, y_2, z_2)$

(i) $T(u) = T(x_1, y_1, z_1)$ $\quad T(v) = T(x_2, y_2, z_2)$

$= (x_1 - y_1, x_1 + z_1)$ $\quad = (x_2 - y_2, x_2 + z_2)$

$T(u) + T(v) = (x_1 - y_1, x_1 + z_1) + (x_2 - y_2, x_2 + z_2)$

$= \left((x_1 - y_1) + (x_2 - y_2) , (x_1 + z_1) + (x_2 + z_2) \right)$

$= \left((x_1 + x_2) - (y_1 + y_2) , (x_1 + x_2) + (z_1 + z_2) \right)$ —(1)

$U + V = (x_1, y_1, z_1) + (x_2, y_2, z_2)$

$= (x_1 + x_2 , y_1 + y_2 , z_1 + z_2)$

$T(U+V) = T(x_1 + x_2, y_1 + y_2, z_1 + z_2)$

$= \left((x_1 + x_2) - (y_1 + y_2) , (x_1 + x_2) + (z_1 + z_2) \right)$ —(2)

From (1) & (2),

$T(U+V) = T(U) + T(V)$

ii) $\det \quad cU = (cx_1, cy_1, cz_1)$

$T(cU) = T(cx_1, cy_1, cz_1)$

$= (cx_1 - cy_1 , cx_1 + cz_1)$

$= (c(x_1 - y_1) , c(x_1 + z_1))$

$= c(x_1 - y_1 , x_1 + z_1)$

$= cT(U)$

3. $\det T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by $T(x, y) = (x + y, x)$

Show that T is a L.T

$\Rightarrow \det \quad U = (x_1, y_1) \qquad V = (x_2, y_2)$

(i) $T(U) = T(x_1, y_1) \qquad T(V) = T(x_2, y_2)$

$T(U+V)$

From (1)

$T(U+V)$

(ii) $\det \quad c$

$T(cU$

4. Show

$(2x_1 -$

$= (x_1+y_1, x_1)$ $= (x_2+y_2, x_2)$

$T(U) + T(V) = (x_1+y_1, x_1) + (x_2+y_2, x_2)$

$= ((x_1+y_1)+(x_2+y_2), (x_1+x_2))$

$= ((x_1+x_2)+(y_1+y_2), (x_1+x_2))$ — (1)

$U+V = (x_1, y_1) + (x_2, y_2)$

$= (x_1+x_2, y_1+y_2)$

$T(U+V) = T(x_1+x_2, y_1+y_2)$

$= T((x_1+x_2)+(y_1+y_2), (x_1+x_2))$ — (2)

From (1) and (2)

$T(U+V) = T(U) + T(V)$

(ii) $cU = c(x_1, y_1)$

$= (cx_1, cy_1)$

$T(cU) = T(cx_1, cy_1)$

$= ((cx_1+cy_1), cx_1)$

$= (c(x_1+y_1), cx_1)$

$= c(x_1+y_1, x_1)$

$= cT(U)$

4. Show that $T: R^3 \to R^3$ defined by $T(x_1, x_2, x_3) = (2x_1 - 3x_2, x_1+4, 5x_3)$ is not a L.T.

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$\Rightarrow$ $\because$ $T(0$ [redacted] $,0,0) \neq 0$

It is not a Linear Transformation.

III. Check whether the mapping $\mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x,y) = (x^2, x+y, 8x-y)$ is a L.T or not.

$\Rightarrow$ $u = (2,4)$ $\quad$ $v = (1,2)$

$T(u) = (4,6,-2)$ $\quad$ $T(v) = (1,3,-1)$

$u+v = (2,4) + (1,2)$ $\quad$ $T(u)+T(v) = (4,6,-2) +$

$= (3,6)$ $\quad$ $(1,3,-1)$

$= (5,9,-3)$

$T(u+v) = T(3,6)$

$= (9,9,-3)$

$\therefore$ $T(u+v) \neq T(u) + T(v)$

Matrix of Linear Transformation:

→ Let v and w be two vector spaces with the dimension m and n respectively. Let $x = \{x_1, x_2, \ldots, x_n\}$ and $y = \{y_1, y_2, \ldots, y_n\}$ are the basis of v and w respectively. If $T: v \rightarrow w$ is a L.T, then $T(x_1)$, $T(x_2), \ldots, T(x_n) \in w$ and each one

of these can be expressed as L.C of

$y_1, y_2, \ldots, y_n$ i.e...

$$T(x_1) = a_1 y_1 + a_2 y_2 + \cdots + a_n y_n$$

$$T(x_2) = a_2 y_1 + a_2 y_2 + \cdots + a_m y_n$$

$$\vdots$$

$$T(x_n) = a_{m_1} y_1 + \cdots + a_{m_n} y_n$$

If A is a coefficient matrix associated with R.H.S then ~~trace~~ $(A)^T$ is called matrix of L.T w.r.t the basis x and y denoted by

$$A^T = \begin{bmatrix} a_{11}, a_{21}, \ldots, a_{m1} \\ a_{12}, a_{22}, \ldots, a_{m2} \\ \vdots \\ a_{2n}, a_{2n}, \ldots, a_{mn} \end{bmatrix}$$

In short $T: \mathbb{R}^m \longrightarrow \mathbb{R}^n$ is a linear transformation then there exist a matrix of mxn dimension such that $T(x) = A^T x$ where A^T - matrix of L.T

NOTE:

- The standard basis of $\mathbb{R}^2$ are $\{(1,0),(0,1)\}$ and $\mathbb{R}^3 = \{(1,0,0),(0,1,0),(0,0,1)\}$ and $\mathbb{R}^4 = \{$

$(1,0,0,0),(0,1,0,0),(0,0,1,0),(0,0,0,1)\}$

- If the basis of V and W are not given take standard basis of V as x and W as 1

IV Consider a linear mapping from $T: \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x,y,z) = (2x+y, y-z)$. Find matrix of linear transformation.

⇒ let the basis of V be $x = \{\underbrace{(1,0,0)}_{e_1}, \underbrace{(0,1,0)}_{e_2}, \underbrace{(0,0,1)}_{e_3}\}$

w.k.T, $T(x,y,z) = (2x+y, y-z)$

Then $T(e_1) = T(1,0,0) = (2,0)$

$T(e_2) = T(0,1,0) = (1,-1)$

$T(e_3) = T(0,0,1) = (0,-1)$

The vectors $T(e_1), T(e_2), T(e_3) \in W = \mathbb{R}^2$

$$T(e_1) = c_1\begin{bmatrix}1\\0\end{bmatrix} + c_2\begin{bmatrix}0\\1\end{bmatrix}$$

$$\begin{bmatrix}2\\0\end{bmatrix} = c_1\begin{bmatrix}1\\0\end{bmatrix} + c_2\begin{bmatrix}0\\1\end{bmatrix}$$

where $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$ are standard basis of $\mathbb{R}^2$

,1,0),(0,0,0,1)}

are not given

V as x and W

om $T: \mathbb{R}^3 \to \mathbb{R}^2$

$, y-z)$ · Find

ion ·

$0,0),(0,1,0),(0,0,1)\}$

e_1 e_2 e_3

z)

2,0)

,1)

-1)

$=\mathbb{R}^2$

$\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$

andard basis

2

$2 = c_1$

$0 = c_2$

$\Rightarrow$ $c_1 = 2$ & $c_2 = 0$

$\therefore$ The matrix of L.T is $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$

V ← Find the matrix of L.T $T: \mathbb{R}^3 \to \mathbb{R}^4$ defined by $T(x,y,z) = (x+y, 2x+y+z, x+3y-z, 3x+z)$

$\Rightarrow$ Let the basis of $\mathbb{R}^3$ are $x = \{(1,0,0), (0,1,0), (0,0,1)\}$ and $y = \{(1,0,0,0), (0,1,0,0), (0,0,1,0), (0,0,0,1)\}$.

$\Rightarrow$ $T(x_1) = T\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{bmatrix}1\\2\\1\\3\end{bmatrix}$ $\quad T(x_2) = T\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{bmatrix}1\\-1\\3\\0\end{bmatrix}$

$T(x_3) = T\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{bmatrix}0\\1\\-1\\1\end{bmatrix}$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & -1 & 1 \\ 1 & 3 & -1 \\ 3 & 0 & 1 \end{bmatrix}$$

Problems on different basis (other than standard basis)

1. Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(x,y) = (x, -y)$ w.r.t basis $\beta_1 = \{(1,1), (1,0)\}$ and $\beta_2 = \{(2,3), (4,5)\}$

⇒ given the basis of domain are $\{x_1 = (1,1), x_2 = (1,0)\}$.

The basis of co domain are $\{y_1 = (2,3), y_2 = (4,5)\}$

Then $T(x_1) = T(1,1) = (1,-1)$

$T(x_2) = T(1,0) = (1,0)$

⇒ $T(x) = c_1 y_1 + c_2 y_2$

$T(x_1) = (1,-1) = c_1(2,3) + c_2(4,5)$

$1 = 2c_1 + 4c_2$

$-1 = 3$

⇒ $c_1 = -9/2$

$T(x_2) = (1$

$\therefore A = \begin{bmatrix} - \\ - \end{bmatrix}$

$\therefore A^T =$

2. Find

$T(x$

and

⇒ Giv

ar

$$-1 = 3C_1 + 5C_2$$

$$\Rightarrow C_1 = -9/2 \text{ and } C_2 = 5/2$$

$$T(x_2) = (1,0) = C_3(2,3) + C_4(4,5)$$

$$1 = 2C_3 + 4C_4$$

$$0 = 3C_3 + 5C_4$$

$$C_3 = -5/2 \text{ and } C_4 = 3/2$$

$$\therefore A = \begin{bmatrix} -9/2 & 5/2 \\ -5/2 & 3/2 \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} -9/2 & -5/2 \\ 5/2 & 3/2 \end{bmatrix}$$

2. Find the matrix of L.T $T: V_2 \to V_3$ defined by $T(x,y) = (x+y, x, 3x-y)$ wit basis $\beta_1 = \{(1,1), (2,1)\}$ and $\beta_2 = \{(1,1,1), (1,1,0), (1,0,0)\}$

$\Rightarrow$ Given the basis of Domain are $x = \left\{ x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}$

and $y = \left\{ y_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, y_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, y_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+y \\ x \\ 3x-y \end{bmatrix}$$

$$T(x_1) = T\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

$$T(x_2) = T\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 8 \end{bmatrix}$$

$$T(x_1) = c_1 y_1 + c_2 y_2 + c_3 y_2$$

$$\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$c_1 + c_2 + c_3 = 2$$

$$c_1 + c_2 = 1$$

$$c_3 = 2$$

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\vec{v}(x_2) = \begin{bmatrix} 4 \\ 3 \\ 8 \end{bmatrix} = c_4 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_5 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_6 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$c_4 + c_5 + c_6 = 4$$

$$c_4 + c_5 = 3$$

$$c_4 = 8$$

$$\begin{bmatrix} c_4 \\ c_5 \\ c_6 \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 1 \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} 2 & 8 \\ -1 & -5 \\ 1 & 1 \end{bmatrix} \quad A^T = \begin{bmatrix} 2 & -1 & 1 \\ 8 & -5 & 1 \end{bmatrix}$$

Nullspace (kernel) & Rangespace (Image) of L.T:

→ Let $T: V \to W$ be a linear transformation then $\ker(T)$ is the set of all vectors in V that mapped to $\vec{0}$ in W i.e....

$$\ker(T) = \{ u \in V \mid T(u) = 0 \}$$

→ The Rangespace is set of all vectors in W that are the images of vectors in V under T denoted by $I_m(T)$ or $\text{Range}(T)$ so,

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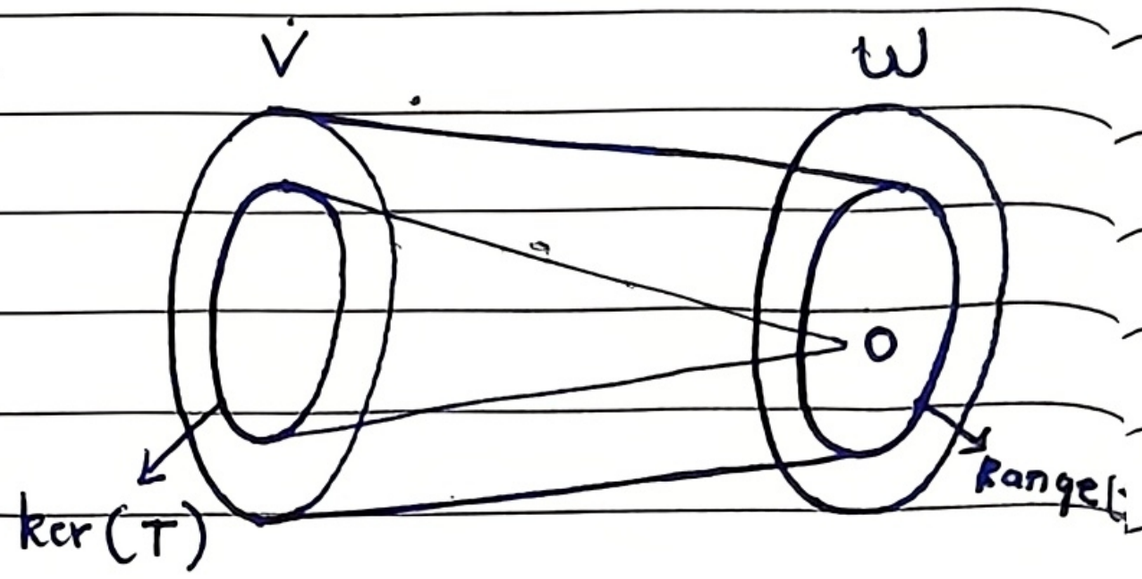
* $I_m(T) = \text{Range}(T) = \{T(x) / x \in V\}$



NOTE

- If $T: V \to W$ is a L.T then $\ker(T)$ is a subspace of V and Range of T is a subspace of W

1. If $T: R^3 \to R^2$ defined by $T(x,y,z) = (x+y, y-z)$ find $\ker(T)$ and $\text{Range}(T)$.

→ To find $\ker(T)$ we have $T(x) = 0$ let $U \in R^3$ and $U = (x,y,z)$ Then $T(U) = 0$ i.e...

$(x+y, y-z) = (0,0)$

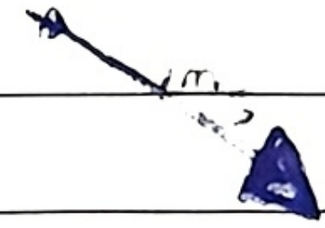
$x+y = 0$ $\quad y-z = 0$

$x = -y$ // $\quad y = +z$

$x = -z$ //

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -z \\ z \\ z \end{bmatrix} :$$

$$= z \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$



$\therefore$ since there are infinite vectors in $\mathbb{R}^3$ so Range(T)= $\mathbb{R}^2$

NOTE:

Ker(T) of kernel & Image of a L.T in terms of matrix representation:

Let $T: V \to W$ be a L.T it can be represented by a matrix A such that $T(u) = Au$ where $u \in V$

We have $\ker(T) = \{u \in V / T(u) = 0\}$ Then $\ker(T) = \{u \in V / Au = 0\}$ which is a nullspace of A.

$\therefore$ Ker(T) = nullspace (A). O

Also since this set $\{Au / u \in V\}$ consists of all the

possible linear combination of columns of A i.e.. exactly the columnspace of A. Thus (Image of T) Im(T) = Columspace (A).

Rank Nullity Theorem:

Let V and W be two vector spaces with L.T $T: V \to W$ then ~~dimension of Range~~

$$\dim(\text{Range}(T)) + \dim(\ker(T)) = \dim(V)$$

OR

Rank(T) + Nullity (T) = dim (V) is called Rank Nullity Theorem where ker(T)-number of vectors in the basis of null space i.e.... dim (ker(T)) = dim (Nullspace) = Nullity (T) and Range (T) - number of vectors in the basis of Image(T) i.e... dim(Range(T)) = dim (Image(T)) = Rank (T).

NOTE

To find dim(ker(T)) & dim(Range(T)) we use following steps:

Obtain the basis of Nullspace (ker(T)) by equating $T(x) = 0$.

Obtain matrix of L.T using standard basis reduce the matrix A into row reduced

echelon form. Obtain the basis of columnspace

3. The obtained basis of Columnspace will be the basis of Rangespace.

✓ The number of elements in the basis of ker(T) and number of elements in basis of Rangespace will be the dimension of ker(T) and Range(T).

1. Let a ~~linear~~ $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x,y,z) = (y-x, y-z)$. Find nullspace and Rangespace. Also prove rank nullity Theorem.

⇒ To find basis of nullspace,

$$v = (x,y,z) \in \mathbb{R}^3$$

$$T(v) = 0$$

$$T(x,y,z) = (0,0)$$

$$(y-x, y-z) = (0,0)$$

$$y - x = 0 \qquad y - z = 0$$

$$x = y \qquad \therefore y = z$$

$$x = z$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} z \\ z \\ z \end{bmatrix} = z \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore$ Nullspace $= \ker(T) = \text{span}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}\right\}$

$\therefore$ Bas

$\therefore$ $\dim(\ker(T)) = \dim(\text{Nullspace}(T)) = 1$

$\therefore$ Dir

To obtain the matrix of L.T, Consider the standard basis,

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di

$$X = \left\{\begin{bmatrix}1\\0\\0\end{bmatrix}\begin{bmatrix}0\\1\\0\end{bmatrix}\begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$$

$C_1 \quad C_2 \quad C_3$

$\therefore$

$$A = TC_1 + TC_2 + TC_3$$

2.

$$T(C_1) = T\begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}-1\\0\end{bmatrix}$$

$$T(C_2) = T\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$$

$$T(C_3) = T\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\-1\end{bmatrix}$$

$\therefore$ $$A = \begin{bmatrix}-1 & 1 & 0\\ 0 & 1 & -1\end{bmatrix}$$

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∴ Basis of columspace = span $\left\{\begin{bmatrix}-1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\right\}$

∴ Dim(Range(T)) = 2

The Rank nullity theorem is dim(Range(T)) + dim(ker(T)) = dim(V) = $\mathbb{R}^3$

∴ 2 + 1 = 3 = $\mathbb{R}^3$ = dim(V)

Hence Proved //

2. Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ is a LT defined by $T(x,y,z) = (x, 0, x+z-y)$ Verify Rank nullity Theorem.

⇒ The Rank nullity Theorem is

Rank(T) + Nullity(T) = dim(V)

w.k.T $T(x) = 0$

$T(x,y,z) = 0$

$(x, 0, x-y+z) = (0, 0, 0)$

$x = 0$ $x - y + z = 0$

$y = z$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ z \\ z \end{bmatrix}$$

$$= z \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

nullspace = span $\left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

dim (nullspace) = dim (nullity (T)) = 1

NOTE

⇒ To find the basis of Range space,

1. Find the matrix of L.T using standard basis.
2. Convert the matrix of L.T into echelon form
3. The Basis of Columnspace will be the basis of Rangespace

The basis of $\mathbb{R}^3 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$T\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{bmatrix}0\\0\\-1\end{bmatrix}$$

$$T\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

Matrix of L.T, $A = \begin{bmatrix}1 & 0 & 0\\0 & 0 & 0\\1 & -1 & 1\end{bmatrix}$

$R_3 \to R_3 - R_1$

$$\begin{bmatrix}1 & 0 & 0\\0 & 0 & 0\\0 & -1 & 1\end{bmatrix}$$

$R_3 \leftrightarrow R_2$

$$\begin{bmatrix}1 & 0 & 0\\0 & -1 & 1\\0 & 0 & 0\end{bmatrix}$$

Basis of ~~Range~~ Rangespace = $\left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \right\}$

dim (Rangespace) = dim (Rank(T)) = 2

$\dim(\mathbb{R}^3) = 3$

∴ Hence dim (Rank(T)) + dim (Nullity(T)) = dim (V)

2 Prove: Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be defined by $T(x,y,z) = (x+y, x-y, x+2z)$

⇒ W.K.T $T(x) = 0$

$T(x,y,z) = 0$

$(x+y, x-y, x+2z) = (0,0,0)$

$x+y = 0 \quad x-y = 0 \quad x+2z = 0$

$x = -y$ $\quad x = y$

$x+y = 0$

$x-y = 0$

$2x = 0$

$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ //

$$\text{nullspace} = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

Nullspace contains only a point. Therefore

$$\dim(\text{nullspace}) = \dim(\text{nullity}(T)) = \dim(\ker(T)) = 0$$

Matrix of L.T,

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

$$T\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$T\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$T\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$R_2 \rightarrow R_2 - R_1$

$R_3 \rightarrow R_3 - R_1$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\text{Columnspace} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \right\}$$

$$\dim(\text{Rangespace}(T)) = 3$$

Inner Product (Dot product):

→ Let u and v are two vectors in $\mathbb{R}^n$, Then the inner product is denoted by $\langle u, v \rangle$

If $u = \{u_1, u_2, \ldots, u_n\}$ and $v = \{v_1, v_2, \ldots, v_n\}$

Then $\langle u, v \rangle = u_1v_1 + u_2v_2 + \ldots + u_nv_n$

Orthogonal Vectors:

→ Let u & v are two vectors, we can say that u & v are orthogonal if and only if the $\langle u, v \rangle = 0$

Orthogonal Set:

→ The set of vectors $v_1, v_2, \ldots, v_n$ in an inner product space is called an orthorgonal set if every pair of distinct vectors in this set is orthogonal. i.e., $\langle v_i, v_j \rangle = 0$ where $i \neq j$

Orthogonal Projection:

→ Let y and u be two vectors in $\mathbb{R}^n$ such that $u \neq 0$. We can write the vector y as the sum of two vectors i.e., $y = \hat{y} + z$ where $\hat{y} = \alpha u$ is a scalar multiple of u (any point on span of u). It lies in the direction of u. A vector z is orthogonal

to u i.e $z \cdot u = 0$. The vector $\hat{y}$ is called Ormogonal projection of y on to u. To find $\hat{y}$ we write $z = y - \hat{y}$, $z \cdot u = 0$, $\hat{y} = \alpha u$

(1) (2)

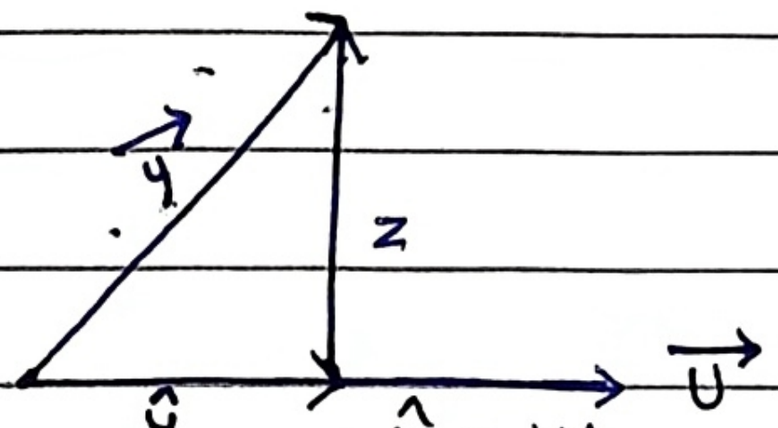
$\therefore$ W.K.T $z = y - \alpha u$

sub (1)

$(y - \alpha u) \cdot u = 0$

$(yu) - \alpha(u \cdot u) = 0$

$\alpha = \left(\frac{y \cdot u}{u \cdot u}\right)$



I Compute the orthogonal projection of $\begin{bmatrix} 1 \\ 7 \end{bmatrix}$ on to the line $\begin{bmatrix} -4 \\ 2 \end{bmatrix}$

$\Rightarrow$ $y = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$ and $u = \begin{bmatrix} -4 \\ 2 \end{bmatrix}$ Then the O.P of

y on to u is given by $\hat{y} = \left[\frac{y \cdot u}{u \cdot u}\right] u$

$y \cdot u = \begin{bmatrix} 1 \\ 7 \end{bmatrix} \cdot \begin{bmatrix} -4 \\ 2 \end{bmatrix} = -4 + 14 = 10$ //

$$u \cdot u = \begin{bmatrix} -4 \\ 2 \end{bmatrix} \begin{bmatrix} -4 \\ 2 \end{bmatrix} = 16 + 4 = 20$$

$$\left[\frac{y \cdot u}{u \cdot u}\right] = \frac{10}{20} = \frac{1}{2} //$$

$$\therefore \hat{y} = \frac{1}{2}\begin{bmatrix} -4 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} //$$

II det $y = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ and $u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ Find the O.P of y on to u and then write y as sum of two orthogonal vectors. one in span of u and another one is orthogonal to u.

$$\hat{y} = \left[\frac{y \cdot u}{u \cdot u}\right] u$$

$$y \cdot u = \begin{bmatrix} 7 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 28 + 12 = 40$$

$$u \cdot u = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 16 + 4 = 20$$

$$\left[\frac{y \cdot u}{u \cdot u}\right] = \frac{40}{20} = 2$$

$$\hat{y} = 2\begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} //$$

$$z = y - \hat{y} = \begin{bmatrix} 7 \\ 6 \end{bmatrix} - \begin{bmatrix} 8 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$y = \hat{y} + z$$

$$= \begin{bmatrix} 8 \\ 4 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix} //$$

$$zu = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 0 //$$

$\hat{y} = \frac{y \cdot u}{u \cdot u} u$

III $y = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $u = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$

$$\hat{y} = \left[\frac{y \cdot u}{u \cdot u}\right] u$$

$$y \cdot u = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ -7 \end{bmatrix} = 8 - 21 = -13$$

$$U \cdot U = \begin{bmatrix} 4 \\ -7 \end{bmatrix} \begin{bmatrix} 4 \\ -7 \end{bmatrix} = 16 + 49 = 65$$

$$\left[\frac{y \cdot u}{u \cdot u}\right] = \frac{-131}{65} = \frac{-1}{5}$$

$$\hat{y} = \frac{-1}{5} \begin{bmatrix} 4 \\ -7 \end{bmatrix} = \begin{bmatrix} -4/5 \\ 7/5 \end{bmatrix}$$

$$z = y - \hat{y} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} - \begin{bmatrix} -4/5 \\ 7/5 \end{bmatrix} = \begin{bmatrix} 14/5 \\ 8/5 \end{bmatrix}$$

$$y = \hat{y} + z = \begin{bmatrix} -4/5 \\ 7/5 \end{bmatrix} + \begin{bmatrix} 14/5 \\ 8/5 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$z \cdot U = 0$

Gram schmidt's Orthogonalization process:

→ Let $\{v_1, v_2, \ldots, v_n\}$ be a linearly independent set of vectors in an inner product space.

The G.S.O.P is a method used to convert this set into an orthogonal set of vectors $\{u_1, u_2, \ldots u_n\}$ which is span the same subspace. The vectors are obtained as follows:

$U_2 = V_2$

$U_3 = V_3$

$U_k = V$

Thus proce and by

1. Let

$$U_2 = V_2 - \left(\frac{V_2 \cdot U_1}{U_1 \cdot U_1}\right) U_1$$

$$U_2 = V_2 - \text{Proj}_{U_1}(V_3) - \text{Proj}_{U_2}(V_3)$$

$$= V_3 - \left(\frac{V_3 \cdot U_1}{U_1 U_1}\right) U_1 - \left(\frac{V_3 U_2}{U_2 \cdot U_2}\right) U_2$$

$$U_k = V_k - \text{Proj}_{U_1}(V_k) - \text{Proj}_{U_2}(V_k) \ldots\ldots - \text{Proj}_{U_n}(V_k)$$

Thus the set $\{U_1, U_2, \ldots\ldots, U_n\}$ obtained by this process is an orthogonal set of vectors and it span the same subspace spanned by vectors $\{V_1, V_2, \ldots\ldots V_n\}$.

1. Let $W = \text{span}\{x_1, x_2\}$ where $x1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$ and $x2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Construct an orthogonal basis $\{U_1, U_2\}$ for W using Gram Schmidt's method.

⇒ Let the independent set be $\{x_1, x_2\}$ and its orthogonal set be $\{U_1, U_2\}$.

Hence $u_1 = x_1$

$$u_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$$v_2 = x_2 - \left(\frac{x_2 \cdot u_1}{u_1 \cdot u_1}\right) u_1$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \left\{ \frac{\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}}{\begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}} \right\} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \left\{ \frac{3 + 12}{9 + 36} \right\} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

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$$= \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} \quad \end{bmatrix} \begin{bmatrix} \quad \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

2. $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ $x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ $x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$

$w = \text{span}\{x_1, x_2, x_3\}$. Find orthogonal basis using G.S method.

$\Rightarrow$ det the independent set be $\{x_1, x_2, x_3\}$ and its orthogonal set be $\{u_1, u_2, u_3\}$

$u_1 = x_1$

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$u_2 = x_2 - \left(\frac{x_2 \cdot u_1}{u_1 \cdot u_1} \right) u_1$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \left\{ \frac{\begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}} \right\} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \left\{ \frac{1+1+1+1}{1+1+1} \right\} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$u_3 = x_3 - \left(\frac{x_3 \cdot u_2}{u_2 \cdot u_2} \right) u_2 - \left(\frac{x_3 \cdot u_1}{u_1 \cdot u_1} \right) u_1$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \left\{ \frac{0+0+1+1}{1+1+1+1} \right\} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \left\{ \frac{0+0+1+1}{1+1+1+1} \right\} \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$$= \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$$= \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \left\{ \frac{0+0+1+1}{1+1+1+1} \right\} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \left\{ \frac{0+0+1+1}{1+1+1+1} \right\} \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \left\{ \frac{1/2}{3/4} \right\} \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \left\{ \frac{1}{2} \times \frac{4}{3} \right\} \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} -1/2 \\ 1/6 \\ 1/6 \\ 1/6 \end{bmatrix}$$

$$= \begin{bmatrix} +1/2 \\ -1/6 \\ 5/6 \\ 5/6 \end{bmatrix}$$

wrong

2. $W = \text{span}\{x_1, x_2, x_3\}$

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 2 \\ -3 \\ -4 \end{bmatrix}$$

$$u_1 = x_1$$

$$= \begin{bmatrix} 1 \\ 2 \\ -3 \\ -4 \end{bmatrix}$$

$$u_2 = x_2 - \left(\frac{x_2 u_1}{u_1 u_1}\right) u_1$$

$$= \begin{bmatrix} 1 \\ -1 \\ 2 \\ 2 \end{bmatrix} - \left\{ \frac{\begin{bmatrix} 1 \\ -1 \\ 2 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}} \right\} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -1 \\ 2 \\ 2 \end{bmatrix} - \left\{ \frac{1-1+2+2}{1+1+1+1} \right\} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -1 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$$u_3 = x_3 - \left(\frac{x_3 \cdot u_1}{u_1 \cdot u_1}\right) u_1 - \left(\frac{x_3 u_2}{u_2 u_2}\right) u_2$$

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$$= \begin{bmatrix} 1 \\ 2 \\ -3 \\ -4 \end{bmatrix} - \left\{ \frac{\begin{bmatrix} 1 \\ 2 \\ -3 \\ -4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}} \right\} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} -$$

$$\left\{ \frac{\begin{bmatrix} 1 \\ 2 \\ -3 \\ -4 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}} \right\} \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ -3 \\ -4 \end{bmatrix} - \left\{ \frac{-4}{4} \right\} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \left\{ \frac{+11}{+2} \right\} \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ -3 \\ -4 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ -11 \\ 11/2 \\ 11/2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -4 \\ -5 \end{bmatrix} - \begin{bmatrix} 0 \\ -1 \\ 11/2 \\ 11/2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ -19/2 \\ -21/2 \end{bmatrix}$$

1 $T(v) = 0$

$T(x, y, z) = 0$

$2x - y + z = 0 \Rightarrow y = 2x + z$

$x + y = 0 \Rightarrow x = -y$

$y - 2z = 0 \Rightarrow y = 2z$

$x - y + 2z = 0 \Rightarrow -2y + 2z$

$y = z$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\dim(\ker(T)) = 0$

$$T\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$T\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 1 \\ -1 \end{bmatrix}$$

$$T\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 2 \end{bmatrix}$$

$A = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$

$\approx \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

Basis =

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$$A = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & -2 \\ 1 & -1 & 2 \end{bmatrix} \quad R_2 \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & -1 \\ 0 & 1 & -2 \\ 0 & -1 & 3 \end{bmatrix}$$

$$\approx \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & -1 \\ 0 & 0 & -5 \\ 0 & 0 & 2 \end{bmatrix}$$

dim (Rangespace) = 3

Basis = $\{(2,1,0,1)\ (-1,1,1,-1)\ (1,0,-2,2)\}$

$\dim(V_3) = 3$

3. $T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5x & -2y \\ -x & -2z \end{bmatrix}$

$\beta_1 = \{(1,2,0)\ (0,-1,0),\ (1,-1,1)\}$

$\beta_2 = \{(1,0)\ (2,-1)\}$

$T(r_1) = 1, -1$

$T(r_2) = 2, 0$

$T(r_3) = 7, -3$

$c_1\begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$c_1 + 2c_2 = 1$
$-c_2 = -1$ $\Rightarrow$ $c_1 = -1$, $c_2 = 1$

$c_1 + 2c_2 = 2$
$-c_2 = 0$ $\Rightarrow$ $c_1 = 2$, $c_2 = 0$

$c_1 + 2c_2 = 7$
$-c_2 = -3$ $\Rightarrow$ $c_1 = 1$, $c_2 = 3$

$A = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$

4. $T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3x + 6y - 5z \\ 3x - 10y + 5z \end{bmatrix}$

$\beta_1 = \{(1,1,1)\ (1,2,3)\ (1,0,0)\}$

$\beta_2 = \{(1,1)\ (1,-1)\}$

$T(r_1) = 4, -2$

$T(r_2) = 0, -2$

$T(r_3) = 3, 3$

$c_1\begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$c_1 + c_2 = 4 \quad \Rightarrow \quad c_1 = 1$$
$$c_1 - c_2 = -2 \qquad\qquad c_2 = 3$$

$$c_1 + c_2 = 0 \quad \Rightarrow \quad c_1 = -1$$
$$c_1 - c_2 = -2 \qquad\qquad c_2 = 1$$

$$c_1 + c_2 = 3 \quad \Rightarrow \quad c_1 = 3$$
$$c_1 - c_2 = 3 \qquad\qquad c_2 = 0$$

$$A = \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 0 \end{bmatrix}$$ //

5. Verify RNT for $T: \mathbb{R}^4 \to \mathbb{R}^3$. $T(x,y,z,t)$
$= \{ x-y+z+t, 2x-2y+3z+4t, 3x-3y+4z+5t \}$

$\Rightarrow T(U) = 0$

$T(x,y,z) = (0,0,0)$

$x - y + z + t = 0$
$2x - 2y + 3z + 4t = 0$
$3x - 3y + 4z + 5t = 0$

$z + 2t = 0$
$-x + y + t = 0 \quad \Rightarrow t = x - y$
$2x - 2y + z = 0 \quad \Rightarrow z = 2(y - x)$

$$\begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} x \\ y \\ (y-x)2 \\ (x-y) \end{bmatrix} = \begin{bmatrix} x \\ y \\ 2y-2x \\ x-y \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix}$$

$\dim(\ker(T)) = 2$

Basis of $\mathbb{R}^4 = \left\{ \begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\1\end{bmatrix} \right\}$

$T(x_1) = T\begin{bmatrix}1\\0\\0\\0\end{bmatrix} = \begin{bmatrix}1\\3\\4\end{bmatrix}$

$T(x_2) = T\begin{bmatrix}0\\1\\0\\0\end{bmatrix} = \begin{bmatrix}-1\\-2\\-3\end{bmatrix}$

$T(x_3) = T\begin{bmatrix}0\\0\\1\\0\end{bmatrix} = \begin{bmatrix}1\\3\\4\end{bmatrix}$

$T(x_4) = T\begin{bmatrix}0\\0\\0\\1\end{bmatrix} = \begin{bmatrix}1\\4\\5\end{bmatrix}$

$$A = \begin{bmatrix}1 & -1 & 1 & 1\\ 2 & -2 & 3 & 4\\ 3 & -3 & 4 & 5\end{bmatrix} \overset{R}{\sim} \begin{bmatrix}1 & -1 & 1 & 1\\ 0 & 0 & 1 & 2\\ 0 & 0 & 1 & 2\end{bmatrix}$$

$$\overset{R}{\sim} \begin{bmatrix}1 & -1 & 1 & 1\\ 0 & 0 & 1 & 2\\ 0 & 0 & 0 & 0\end{bmatrix}$$

Basis of $R^4 = \left\{ \begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}1\\3\\4\end{bmatrix} \right\}$

$\dim(\text{Rangespace}) = 2$, $\dim(\mathbb{R}^4) = 4$ $2 + 2 = 4$ //